

Question 1

Given $u(x, 0) = x^2$, we want

$$u_t = u(x, 0)_{xx} = 2$$

Integrating with respect to t , we get $u = 2t + C$. Since $u(x, 0) = x^2$, we take $C = 0$. So we have

$$u(x, t) = x^2 + 2t$$

Question 2

(a) Given $u_t = Du_{xx} + f$. We have u_t meaning the rate of change in temperature of the rod over time, Du_{xx} means the heat flux of the rod, and f means the heat source.

(b) (i) Given $f(x) = 1$ with BCs $u(0, t) = 0$, $u(l, t) = 1$

$$u_t = Du_{xx} + 1$$

Since the steady state solution is time independent, $u_t = 0$

$$Du_{xx} = -1$$

$$u_{xx} = -\frac{1}{D}$$

$$u_x = -\frac{1}{D}x + f(t)$$

$$u = -\frac{x^2}{2D} + ax + b$$

Using BCs $u(0, t) = 0$, $u(l, t) = 1$

$$u(0) = -\frac{0^2}{2D} + a \cdot 0 + b \Rightarrow b = 0$$

$$u(l) = -\frac{l^2}{2D} + al = 0$$

$$\Rightarrow \frac{l^2}{2D} = al$$

$$\frac{l}{2D} = a$$

$$\Rightarrow u(x) = -\frac{x^2}{2D} + \frac{lx}{2D}$$

(ii) Given $f(x) = \begin{cases} 0 & 0 < x < l/2 \\ H & l/2 < x < l \end{cases}$ with BCs $u(0, t) = 0, u(l, t) = 0$

We have $Du_{xx} = 0$ for $0 < x < l/2$

$$Du_{xx} = 0$$

$$Du_x = a$$

$$Du = ax + b$$

$$u = \frac{a}{D}x + \frac{b}{D}$$

and $Du_{xx} = -H$ for $l/2 < x < l$

$$Du_{xx} = -H$$

$$Du_x = -Hx + c$$

$$Du = -\frac{Hx^2}{2} + cx + e$$

$$u = -\frac{Hx^2}{2D} + \frac{c}{D}x + \frac{e}{D}$$

With our BCs, we have

$$u(0) = \frac{a}{D} \cdot 0 + \frac{b}{D} = 0$$

$$\Rightarrow \frac{b}{D} = 0$$

$$u(l) = -\frac{Hl^2}{2D} + \frac{c}{D}l + \frac{e}{D} = 0$$

$$\Rightarrow \frac{e}{D} = \frac{Hl^2}{2D} - \frac{c}{D}l$$

Considering the rod is continuous, we have two more constraints, where heat $u(x)$ and flux $-Du_x$ are continuous at $\frac{l}{2}$. With heat, we have

$$u\left(\frac{l}{2}-\right) = u\left(\frac{l}{2}+\right)$$

$$\frac{al}{2D} = -\frac{Hl^2}{8D} + \frac{cl}{2D} + \frac{Hl^2}{2D} - \frac{cl}{D}$$

$$\frac{al}{2D} = \frac{3Hl^2}{8D} - \frac{cl}{2D}$$

$$\frac{a}{D} = \frac{3Hl}{4D} - \frac{c}{D}$$

With heat flux, we have

$$\begin{aligned} -Du_x\left(\frac{l}{2}-\right) &= -Du_x\left(\frac{l}{2}+\right) \\ u_x\left(\frac{l}{2}-\right) &= u_x\left(\frac{l}{2}+\right) \\ \frac{a}{D} &= -\frac{Hl}{2D} + \frac{c}{D} \end{aligned}$$

Combining the two, we get

$$\begin{aligned} \frac{3Hl}{4D} - \frac{c}{D} &= -\frac{Hl}{2D} + \frac{c}{D} \\ \frac{5Hl}{4D} &= \frac{2c}{D} \\ \frac{5Hl}{8D} &= \frac{c}{D} \\ \Rightarrow \frac{a}{D} &= -\frac{Hl}{2D} + \frac{5Hl}{8D} \\ &= \frac{Hl}{8D} \\ \Rightarrow \frac{e}{D} &= \frac{Hl^2}{2D} - \frac{5Hl^2}{8D} \\ &= -\frac{Hl^2}{8D} \end{aligned}$$

And so our solutions are

$$\begin{cases} u = \frac{Hl}{8D}x & 0 < x < \frac{l}{2} \\ u = -\frac{Hx^2}{2D} + \frac{5Hl}{8D}x - \frac{Hl^2}{8D} & \frac{l}{2} < x < l \end{cases}$$

Question 3